MY DUY HAI ANH

AI Engineer

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Professional Summary

AI-oriented Computer Science student with hands-on experience in Deep Learning and Natural Language Processing (NLP), focusing on sequence-to-sequence models, sequence labeling, and neural machine translation. Proficient in developing, fine-tuning, and evaluating complex sequential architectures (RNNs, LSTMs, Transformers) using PyTorch and Hugging Face Ecosystem. Experienced in building end-to-end NLP pipelines—including advanced tokenization, text preprocessing, and dataset curation—as well as conducting large-scale experiments and optimization.

Education

Hanoi University of Science and Technology **2024 – Present**
Computer Science *Hanoi, Vietnam*

- Relevant coursework: Artificial Intelligence, Machine Learning, Deep Learning, Data Structures and Algorithms, Databases, Computer Networks, and Operating Systems.
- GPA: 3.6/4.0.
- CPA: 3.64/4.0.

Experience

AI Research Trainee **10/2025 – Present**
AI4LIFE Laboratory *Hanoi, Vietnam*

- Conducted experiments on Natural Language Processing (NLP) methods for large-scale text analysis and neural machine translation under the supervision of laboratory researchers.
- Prepared training and evaluation datasets by reviewing tokenization boundaries, correcting label inconsistencies in POS tagging/NER, and converting between different text annotation formats.
- Analyzed model failure cases, including out-of-vocabulary (OOV) tokens, alignment errors, syntax confusion, and fragmented sequence generation, to identify potential areas for improvement.
- Used Linux servers, CUDA, Conda, tmux, and Git to manage experiments and source code.

Projects

Gloss2Text Machine Translation (for Sign Language) Source Code
Python, PyTorch, VnCoreNLP, mBART, BiLSTM, Multitask Learning, Data augmentation

- Developed and optimized deep learning architectures for Sign Language Gloss-to-Text translation, converting symbolic gloss sequences into grammatically correct natural language text.
- Fine-tuned state-of-the-art pretrained Sequence-to-Sequence models, including BARTpho, mT5, mBART-50, and NLLB, adapting cross-lingual models to handle the unique grammar and low-resource constraints of sign language glosses.
- Built robust text processing pipelines to handle sign language specific challenges, including out-of-vocabulary (OOV) glosses, non-manual marker representations, and variable sequence lengths.
- Mitigated hardware limitations during large-scale training on remote Linux servers by implementing memory optimization techniques such as gradient accumulation, mixed-precision training, and model quantization.
- Evaluated translation quality using standard machine translation metrics, conducting rigorous error analysis on BLEU, ROUGE, CHRF, and WER scores to improve syntactic alignment and generation fluency.

Reproducing SALS: LLM-based Sign Language Gloss Translation Source Code
Python, PyTorch, Hugging Face, BARTpho, mT5, LoRA, Semantically Aware Label Smoothing

- Reproduced and adapted the paper "Sign Language Gloss translation using LLMs and Semantically Aware Label Smoothing" to evaluate text generation performance on Vietnamese Gloss and RWTH-PHOENIX-Weather 2014T datasets.
- Curated, cleaned, and standardized parallel text datasets, resolving target-side sentence inconsistencies and man-

aging text normalization tokens (e.g., handling uppercase gloss semantic constraints).

- Implemented advanced sequence processing and cross-lingual embeddings, integrating custom tokenization boundaries, position embeddings, and semantic similarity matrices for text alignment.
- Compared autoregressive and sequence-to-sequence language models, establishing a strong baseline with BARTpho full-finetuning and benchmarking against various configurations of mT5 (Small, Base, and Uppercase Gloss variants).
- Enhanced generation quality and sample efficiency by integrating Parameter-Efficient Fine-Tuning via LoRA and implementing Semantically Aware Label Smoothing (SALS) combined with Cross-Entropy loss.
- Conducted comprehensive evaluations across multiple NLP metrics, achieving a baseline BLEU score of 47.77 (BARTpho) and a reproduced SALS + CE loss BLEU score of 43.75 with LoRA, while monitoring Rouge-1/2/L and CHRF++ scores to analyze generation fluency.

Technical Skills

Programming Languages	Python, Java, C#, C, C++
AI / Machine Learning	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Deep Learning
Natural Language Processing	Sequence Labeling (POS Tagging, NER), Neural Machine Translation (NMT), Sequence-to-Sequence (Seq2Seq)
Tools & Infrastructure	Git, GitHub, Linux, Conda, tmux, CUDA
Databases	MySQL, PostgreSQL (Basic)

Languages

Vietnamese: Native
English: Intermediate; able to read technical documentation and research papers